AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 19: Integral Calculus

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1 Keywords

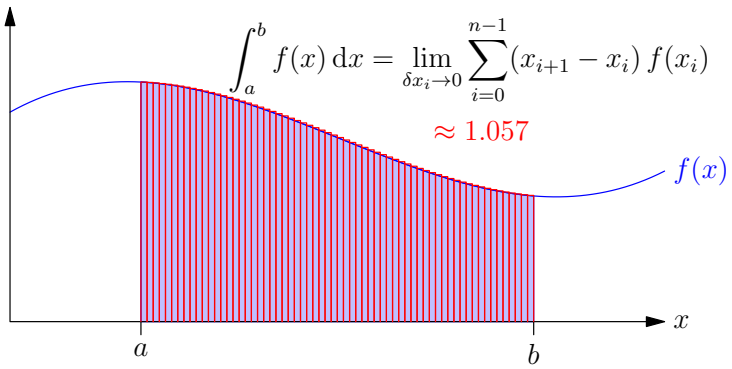
- Riemann integral, fundamental theorem of calculus, integration by parts, substitution, Jacobians, generating functions, gamma function

2 Main Points

2.1 The Riemann Integral

- An integral is the area under a curve. Divide $[a, b]$ into strips at points $a = x_0 < x_1 < \cdots < x_n = b$ and add up the areas of the rectangles:

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\delta x_i \to 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i)\, f(x_i)$$



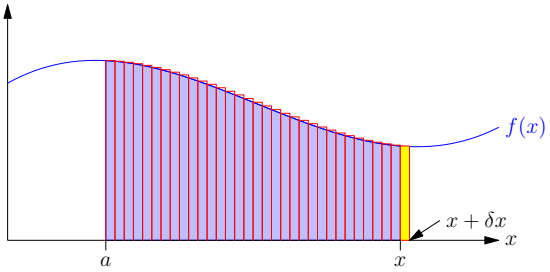
- Integration is linear: splitting the sum,

$$\int_a^b (r\, f(x) + s\, g(x))\,\mathrm{d}x = r \int_a^b f(x)\,\mathrm{d}x + s \int_a^b g(x)\,\mathrm{d}x$$

2.2 The Fundamental Theorem of Calculus

- Let $I(a, x) = \int_a^x f(z)\,\mathrm{d}z$. Extending the upper limit by δx adds a thin strip of area $\delta x\, f(x)$, so

$$\frac{\mathrm{d}I(a,x)}{\mathrm{d}x} = \lim_{\delta x \to 0} \frac{I(a, x+\delta x) - I(a,x)}{\delta x} = f(x)$$



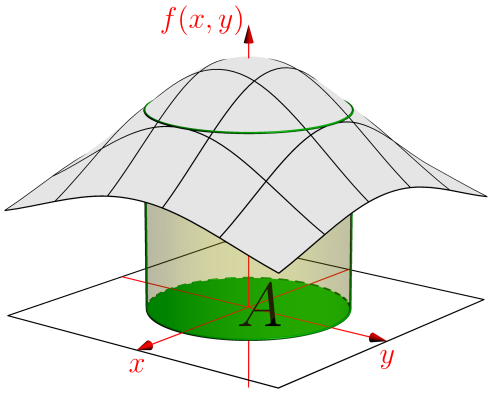
- Conversely, integrating a derivative, the Riemann sum telescopes:

$$\int_a^b \frac{\mathrm{d}f(x)}{\mathrm{d}x}\,\mathrm{d}x = \lim \sum_{i=0}^{n-1} (x_{i+1} - x_i)\,\frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} = \lim \sum_{i=0}^{n-1} \left(f(x_{i+1}) - f(x_i)\right)$$
$$= f(x_n) - f(x_0) = f(b) - f(a)$$

- So integration is an *anti-derivative*: it undoes differentiation

2.3 Indefinite and Multiple Integrals

- An *indefinite integral* is a function $F(x) = \int f(x)\,\mathrm{d}x$ with $F'(x) = f(x)$. Since $F(x)$, $F(x)+1$, $F(x)+\pi, \ldots$ all have the same derivative, F is only defined up to an additive constant
- Definite integrals follow from it: $\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$
- For functions of several variables we integrate over regions, e.g. $\iint_A f(x,y)\,\mathrm{d}x\,\mathrm{d}y$. We usually write a single integral sign, $\int f(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}$, for $\int \cdots \int f(x_1, \ldots, x_n)\,\mathrm{d}x_1 \cdots \mathrm{d}x_n$



2.4 Performing Integrals

- The basic method is to recognise an anti-derivative. Since $\mathrm{d}x^n/\mathrm{d}x = n\,x^{n-1}$,

$$\int x^{n-1}\,\mathrm{d}x = \frac{x^n}{n} + c \qquad (n \neq 0), \qquad \int_a^b x^{n-1}\,\mathrm{d}x = \frac{b^n - a^n}{n}$$

- The product and chain rules let us differentiate almost anything, but there are no general rules for $\int f(x)\,g(x)\,\mathrm{d}x$ or $\int f(g(x))\,\mathrm{d}x$. Unlike differentiation there is no small parameter to expand in. *In general integration is hard*

2.5 Integration by Parts

- Integrating the product rule $\mathrm{d}(f\,g)/\mathrm{d}x = f'\,g + f\,g'$ from a to b and rearranging,

$$\int_a^b f(x)\,\frac{\mathrm{d}g(x)}{\mathrm{d}x}\,\mathrm{d}x = \Big[f(x)\,g(x)\Big]_a^b - \int_a^b \frac{\mathrm{d}f(x)}{\mathrm{d}x}\,g(x)\,\mathrm{d}x$$

  Whether this helps depends on whether the new integral is easier

- **Example: the factorial function.** Let $\Pi(z) = \int_0^\infty x^z\,\mathrm{e}^{-x}\,\mathrm{d}x$. Taking $g(x) = -\mathrm{e}^{-x}$,

$$\Pi(z) = \Big[-x^z\,\mathrm{e}^{-x}\Big]_0^\infty + \int_0^\infty z\,x^{z-1}\,\mathrm{e}^{-x}\,\mathrm{d}x = z\,\Pi(z-1)$$

  since the boundary term vanishes for $z > 0$

- $\Pi(0) = \int_0^\infty \mathrm{e}^{-x}\,\mathrm{d}x = \Big[-\mathrm{e}^{-x}\Big]_0^\infty = 1$, so for integers $\Pi(n) = n\,(n-1)\cdots 1 = n!$

2.6 Differentiating Through the Integral

- Sometimes we can do an integral by differentiating a related, easier one. The moments of a random variable are $M_n = \mathbb{E}[X^n] = \int x^n \, f_X(x) \, \mathrm{d}x$
- Define the *moment generating function*

$$Z(\ell) = \mathbb{E}\left[\mathrm{e}^{\ell X}\right] = \int_{-\infty}^{\infty} \mathrm{e}^{\ell x} \, f_X(x) \, \mathrm{d}x$$

- Differentiating n times under the integral and setting $\ell = 0$,

$$\left.\frac{\mathrm{d}^n Z(\ell)}{\mathrm{d}\ell^n}\right|_{\ell=0} = \int_{-\infty}^{\infty} x^n \, f_X(x) \, \mathrm{d}x = M_n$$

2.7 The Cumulant Generating Function

- Expanding $\mathrm{e}^{\ell x} = 1 + \ell \, x + \frac{1}{2}\ell^2 x^2 + \cdots$ inside the integral,

$$Z(\ell) = 1 + \ell \, M_1 + \frac{1}{2}\ell^2 M_2 + \frac{1}{3!}\ell^3 M_3 + \cdots$$

- Taking the logarithm, using $\log(1+\epsilon) = \epsilon - \frac{1}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 - \cdots$, gives the *cumulant generating function*

$$G(\ell) = \log(Z(\ell)) = \ell \, M_1 + \frac{1}{2}\ell^2 \left(M_2 - M_1^2\right) + \frac{1}{3!}\ell^3 \left(M_3 - 3\, M_2 \, M_1 + 2\, M_1^3\right) + \cdots$$

- The *cumulants* are $\kappa_n = G^{(n)}(0)$: κ_1 is the mean, κ_2 the variance and κ_3 is related to the skewness

2.8 Special Functions

- Some integrals have no closed form in terms of elementary functions. $\Pi(z)$ is one: $\Pi(n) = n!$ for integers, but for general z it is a new function in its own right
- History has left us with the *gamma function* instead,

$$\Gamma(z) = \int_0^{\infty} x^{z-1} \, \mathrm{e}^{-x} \, \mathrm{d}x = \Pi(z-1)$$

- Other special functions defined by integrals include the Bessel, Airy, hypergeometric, elliptic and error functions
- Integration is sometimes easier in the complex plane (largely due to Cauchy), and there is an algorithm (the Risch algorithm) that decides whether many functions have elementary integrals. It is complicated enough that it is only available inside symbolic packages such as Mathematica. Both are beyond this course

2.9 Substitution

- Changing variable from x to u, with $x = x(u)$, the Riemann sum becomes

$$\int_a^b f(x) \, \mathrm{d}x = \int_{u(a)}^{u(b)} f(x(u)) \, \frac{\mathrm{d}x(u)}{\mathrm{d}u} \, \mathrm{d}u$$

  where $u(x)$ is the inverse of $x(u)$

- **Example.** For $I(n) = \int_0^\infty x^n \, \mathrm{e}^{-x^2/2} \, \mathrm{d}x$, put $u = x^2/2$, so $x = \sqrt{2u}$, $\mathrm{d}x/\mathrm{d}u = 1/\sqrt{2u}$, and the limits stay 0 and ∞:

$$I(n) = \int_0^\infty (2u)^{n/2} \, \mathrm{e}^{-u} \, \frac{\mathrm{d}u}{\sqrt{2u}} = 2^{\frac{n-1}{2}} \int_0^\infty u^{\frac{n-1}{2}} \, \mathrm{e}^{-u} \, \mathrm{d}u = 2^{\frac{n-1}{2}} \, \Pi\left(\frac{n-1}{2}\right)$$

- So $I(1) = 1$, $I(3) = 2 \times 1! = 2$ and $I(5) = 2^2 \times 2! = 8$. Even n need Π at half integers, e.g. $I(0) = \Pi(-1/2)/\sqrt{2}$, and $I(2) = \sqrt{2}\,\Pi(1/2) = \Pi(-1/2)/\sqrt{2}$ using $\Pi(1/2) = \frac{1}{2}\Pi(-1/2)$

- **A recipe.** (1) Choose $u(x)$, often an "inner" function whose derivative also appears in the integrand; (2) write $\mathrm{d}u = u'(x) \, \mathrm{d}x$ and replace every x; (3) change the limits; (4) integrate in u

- **Example.** With $u = 1 + x^2$, $\mathrm{d}u = 2x \, \mathrm{d}x$ and limits $u : 1 \to 2$,

$$\int_0^1 2x \, (1 + x^2)^3 \, \mathrm{d}x = \int_1^2 u^3 \, \mathrm{d}u = \frac{16 - 1}{4} = \frac{15}{4}$$

- **The limits can swap over.** With $u = \cos(x)$, $\mathrm{d}u = -\sin(x) \, \mathrm{d}x$ and $u : 1 \to 0$,

$$\int_0^{\pi/2} \sin(x) \cos^2(x) \, \mathrm{d}x = \int_1^0 -u^2 \, \mathrm{d}u = \int_0^1 u^2 \, \mathrm{d}u = \frac{1}{3}$$

- **A linear change of variable stretches** $\mathrm{d}x$**.** With $z = (x - \mu)/\sigma$, $\mathrm{d}x = \sigma \, \mathrm{d}z$; the factor σ is the one-dimensional Jacobian determinant

2.10 The Gamma Distribution

- In the lecture on Bayesian inference we met the *Gamma distribution* $\mathrm{Gamma}(x|a, b) = c \, x^{a-1} \mathrm{e}^{-b\,x}$ for $x \geq 0$. To find c, substitute $z = b\,x$ (so $\mathrm{d}x = \mathrm{d}z/b$, and the limits stay 0 and ∞):

$$c \int_0^\infty x^{a-1} \mathrm{e}^{-b\,x} \, \mathrm{d}x = \frac{c}{b^a} \int_0^\infty z^{a-1} \mathrm{e}^{-z} \, \mathrm{d}z = \frac{c\,\Gamma(a)}{b^a} = 1 \quad \Rightarrow \quad c = \frac{b^a}{\Gamma(a)} = \frac{b^a}{\Pi(a-1)}$$

- In terms of $z = b\,x$ every Gamma distribution becomes $\mathrm{Gamma}(z|a, 1)$, so b only sets the scale and a the shape (Figure 1): for $a < 1$ the density is infinite at 0, $a = 1$ is the exponential distribution, and for $a > 1$ there is a peak at $z = a - 1$

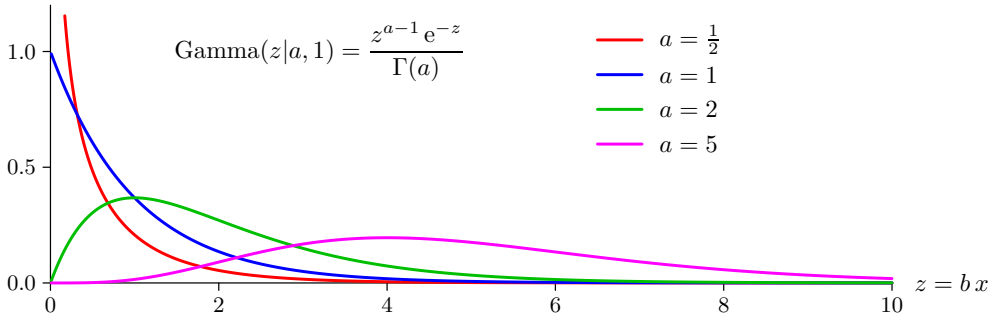


Figure 1: Gamma distributions for $a = \frac{1}{2}, 1, 2, 5$ in terms of \$z = b\,x\$

- **Lengths of Gaussian vectors.** Let $\boldsymbol{x} \in \mathbb{R}^n$ with $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The density $(2\pi)^{-n/2} \mathrm{e}^{-r^2/2}$ depends only on $r = \|\boldsymbol{x}\|$. It is largest at $\boldsymbol{x} = \mathbf{0}$, but the volume of a thin shell of radius r and thickness $\mathrm{d}r$ (Figure 2) grows as $r^{n-1}\mathrm{d}r$, so

$$f_R(r) = \frac{r^{n-1} \mathrm{e}^{-r^2/2}}{I(n-1)}, \qquad I(n-1) = \int_0^\infty r^{n-1} \mathrm{e}^{-r^2/2} \mathrm{d}r = 2^{\frac{n}{2}-1} \, \Gamma\left(\frac{n}{2}\right)$$

using $I(n)$ from the substitution example

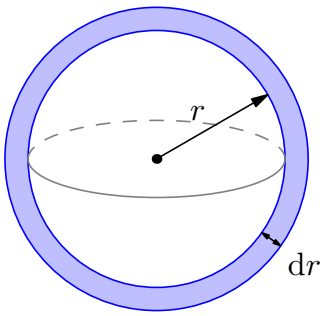


Figure 2: A thin shell of radius r and thickness $\mathrm{d}r$

- Maximising $(n-1)\log(r) - r^2/2$ puts the peak at $r = \sqrt{n-1}$. In high dimensions almost all the probability lies in a thin shell at $r \approx \sqrt{n}$, far from the most probable point $\boldsymbol{x} = \mathbf{0}$
- Changing variables to $s = r^2$ gives $f_S(s) = \mathrm{Gamma}(s|\frac{n}{2}, \frac{1}{2})$ (see the exercises): the *chi-squared* distribution χ^2_n with n degrees of freedom
- **Gamma distributions everywhere.**
  - For m samples from a normal distribution the sample variance satisfies $(m-1)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{m-1}$, which is what lies behind the t-test
  - *Maxwell–Boltzmann*: each velocity component of a gas molecule of mass M at temperature T is $\mathcal{N}(0, kT/M)$. The speed is the $n = 3$ case, $f(v) \propto v^2 \mathrm{e}^{-Mv^2/(2kT)}$, and the kinetic energy $\frac{1}{2}Mv^2 = \frac{kT}{2}\chi^2_3$ is $\mathrm{Gamma}(\frac{3}{2}, \frac{1}{kT})$
  - *Waiting times*: $a = 1$ is the exponential distribution, and the time to the a-th event of a Poisson process of rate b is $\mathrm{Gamma}(a, b)$
  - the Gamma distribution is the conjugate prior for a Poisson rate

2.11 Changing Variables in Many Dimensions

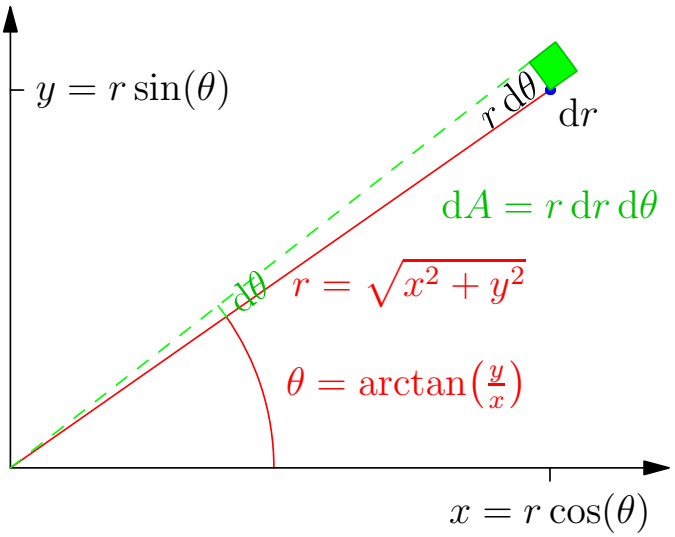
- Changing variables $\boldsymbol{x} \to \boldsymbol{u}$, volume elements are rescaled by the determinant of the Jacobian:

$$\int f(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} = \int f(\boldsymbol{x}(\boldsymbol{u}))\,|\det(\mathbf{J})|\,\mathrm{d}\boldsymbol{u}, \qquad J_{ij} = \frac{\partial x_i}{\partial u_j}$$

- **Polar coordinates.** With $x = r\cos(\theta)$ and $y = r\sin(\theta)$,

$$|\det(\mathbf{J})| = \left|\det\begin{pmatrix}\cos(\theta) & -r\sin(\theta)\\ \sin(\theta) & r\cos(\theta)\end{pmatrix}\right| = r\left(\cos^2(\theta) + \sin^2(\theta)\right) = r$$

so $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$: a small patch at radius r has sides $\mathrm{d}r$ and $r\,\mathrm{d}\theta$



- Gaussian integrals, which use these changes of variables, are the subject of the next lecture (see the Gaussian Integrals notes)

3 Exercises

3.1 Integration by parts

- Evaluate $\int_0^1 x\,\mathrm{e}^x\,\mathrm{d}x$

3.2 Substitution

- Evaluate $\int_0^\infty x\,\mathrm{e}^{-x^2}\,\mathrm{d}x$

3.3 Mean absolute value of a normal

- For a standard normal Z, use $I(1)$ to find $\mathbb{E}[|Z|]$

3.4 From radius to chi-squared

- Starting from $f_R(r) = r^{n-1}\mathrm{e}^{-r^2/2}/I(n-1)$, use the change of variables $s = r^2$ to show that $f_S(s) = \mathrm{Gamma}(s|\frac{n}{2},\frac{1}{2})$.

3.5 The surface area of a sphere in n dimensions (advanced)

- The unit sphere in $\mathbb{R}^n$ is $S^{n-1} = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\| = 1\}$. It is called the $(n-1)$-sphere because it is an $(n-1)$-dimensional surface: S^1 is the circle and S^2 the ordinary sphere. We write $|S^{n-1}|$ for its surface area (you will also see ω_{n-1}, S_{n-1} or Ω_n, the total solid angle)

- The normal density $\mathcal{N}(\boldsymbol{x}|\mathbf{0},\mathbf{I}) = (2\pi)^{-n/2}\,\mathrm{e}^{-\|\boldsymbol{x}\|^2/2}$ integrates to 1. In spherical coordinates the volume element is $\mathrm{d}\boldsymbol{x} = r^{n-1}\,\mathrm{d}r\,\mathrm{d}\Omega$, where $r = \|\boldsymbol{x}\|$ and $\mathrm{d}\Omega$ is the element of area on S^{n-1}. Use this to show that
$$|S^{n-1}| = \frac{2\,\pi^{n/2}}{\Gamma(n/2)}$$
and check it for $n = 2$ and $n = 3$. (Hint: the radial integral is $I(n-1)$ from the lecture.)

- Since a ball of radius R is a stack of shells, its volume is $V_n(R) = \int_0^R |S^{n-1}|\,r^{n-1}\mathrm{d}r$. Find the volume V_n of the unit ball. For which n is it largest, and what happens as $n \to \infty$?

4 Answers

4.1 Integration by parts

- Take $f(x) = x$ and $g(x) = \mathrm{e}^x$:
$$\int_0^1 x\,\mathrm{e}^x\,\mathrm{d}x = \Big[x\,\mathrm{e}^x\Big]_0^1 - \int_0^1 \mathrm{e}^x\,\mathrm{d}x = \mathrm{e} - (\mathrm{e}-1) = 1$$

4.2 Substitution

- With $u = x^2$, $\mathrm{d}u = 2\,x\,\mathrm{d}x$:
$$\int_0^\infty x\,\mathrm{e}^{-x^2}\,\mathrm{d}x = \frac{1}{2}\int_0^\infty \mathrm{e}^{-u}\,\mathrm{d}u = \frac{1}{2}$$

4.3 Mean absolute value of a normal

- By symmetry
$$\mathbb{E}[|Z|] = \frac{2}{\sqrt{2\pi}} \int_0^\infty z\, \mathrm{e}^{-z^2/2}\, \mathrm{d}z = \frac{2\, I(1)}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}} \approx 0.798$$

4.4 From radius to chi-squared

- With $r = \sqrt{s}$, $\mathrm{d}r/\mathrm{d}s = 1/(2\sqrt{s})$, so
$$f_S(s) = f_R(\sqrt{s})\, \frac{1}{2\sqrt{s}} = \frac{s^{(n-1)/2}\, \mathrm{e}^{-s/2}}{2\sqrt{s}\; 2^{\frac{n}{2}-1} \Gamma(\frac{n}{2})} = \frac{(1/2)^{n/2}}{\Gamma(\frac{n}{2})}\, s^{\frac{n}{2}-1} \mathrm{e}^{-s/2}$$
which is $\mathrm{Gamma}(s|a,b)$ with $a = \frac{n}{2}$, $b = \frac{1}{2}$

4.5 The surface area of a sphere in n dimensions (advanced)

- The integrand depends only on r, so the integral over directions just gives the area of the unit sphere:
$$1 = \int \mathcal{N}(\boldsymbol{x}|\mathbf{0}, \mathbf{I})\, \mathrm{d}\boldsymbol{x} = (2\pi)^{-n/2}\, |S^{n-1}| \int_0^\infty r^{n-1} \mathrm{e}^{-r^2/2}\, \mathrm{d}r = (2\pi)^{-n/2}\, |S^{n-1}|\; 2^{\frac{n}{2}-1}\, \Gamma\Big(\frac{n}{2}\Big)$$
using $I(n-1) = 2^{\frac{n}{2}-1}\Gamma(\frac{n}{2})$. Hence
$$|S^{n-1}| = \frac{(2\pi)^{n/2}}{2^{\frac{n}{2}-1}\, \Gamma(n/2)} = \frac{2\, \pi^{n/2}}{\Gamma(n/2)}$$
and a sphere of radius R has area $|S^{n-1}|\, R^{n-1}$

- Checks: $n = 2$ gives $2\pi/\Gamma(1) = 2\pi$, the circumference of the unit circle; $n = 3$ gives $2\pi^{3/2}/\Gamma(\frac{3}{2}) = 2\pi^{3/2}/(\frac{1}{2}\sqrt{\pi}) = 4\pi$; and $n = 1$ gives $2\sqrt{\pi}/\Gamma(\frac{1}{2}) = 2$: the "sphere" S^0 is the two points ± 1

- The volume of the unit ball is
$$V_n = |S^{n-1}| \int_0^1 r^{n-1} \mathrm{d}r = \frac{|S^{n-1}|}{n} = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}$$
using $\frac{n}{2}\Gamma(\frac{n}{2}) = \Gamma(\frac{n}{2}+1)$. This gives $V_2 = \pi$ and $V_3 = \frac{4}{3}\pi$. It grows to a maximum $V_5 \approx 5.26$ and then falls ($V_{10} \approx 2.55$, $V_{20} \approx 0.026$): $\Gamma(\frac{n}{2}+1)$ eventually grows much faster than $\pi^{n/2}$, so $V_n \to 0$ as $n \to \infty$. In high dimensions the unit ball occupies a vanishing fraction of the cube $[-1,1]^n$ around it, whose volume is 2^n